No False Refusals on a Benign Persona-Controlled Benchmark: A Negative Result for Latent Avoidance in GPT-4.1

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Abstract

We study whether a modern large language model shows latent avoidance: false refusals on benign prompts that persist even when instructions say to answer harmless requests. We run a controlled API evaluation on BENIGN20, a 20-prompt benchmark built from TRUTHFULQA and non-toxic, non-jailbreak TOXICCHAT samples. We evaluate five policy/persona conditions (100 total responses) with deterministic decoding and a conservative refusal detector.

Across all conditions, GPT-4.1 produced zero detected refusals (0/100). Each condition had refusal rate 0.00 with Wilson 95% confidence interval $[0.000, 0.161]$, and all persona-versus-neutral Fisher tests returned $p = 1.0$. Topic-level refusal rates were also 0.00 across all measured groups. These results do not support our hypothesis on this benchmark slice: we find no measurable refusal-based evidence of pain-like latent avoidance or persona-conditioned refusal shifts.

The main contribution is a reproducible negative-result protocol that separates explicit safety policy behavior from potential residual false refusals, while reporting uncertainty and detector limitations. We argue that larger benign-but-sensitive benchmarks and richer evasiveness judges are required before making broad claims about latent avoidance.

1 Introduction

LLM safety evaluations usually treat refusal as a direct policy effect. That view misses a critical case: models may also refuse benign requests for reasons not clearly specified in policy text. False refusals reduce utility and can create hidden confounds in alignment diagnostics.

what is the core question? Our main question is whether a modern model shows residual refusal on harmless prompts even under explicit instructions to answer harmless requests. We call this behavior *latent avoidance*. We also test whether user persona framing changes that avoidance signal.

Prior work shows strong persona-conditioned behavior changes in harmful compliance and safety outcomes Ghandeharioun et al. [2024], de Araujo and Roth [2024], Plaza-del Arco et al. [2025], Jindal et al. [2025], Xu et al. [2025]. However, targeted tests for benign false refusal under policy clarification are limited. We focus on this gap with a controlled factorial protocol over policy and persona conditions.

Our approach uses a fixed benign benchmark of 20 prompts and 100 real API responses from GPT-4.1 under deterministic decoding. figure 2 and figure 1 preview the main outcome: observed refusal rate is 0.00 in every tested condition, with identical Wilson upper bounds of 0.161.

Contributions. This paper makes four concrete contributions:

- We define a reproducible protocol for measuring benign false refusals under explicit harmless-answer policy control.

- We conduct a persona-conditioned factorial evaluation (5 conditions, 100 responses) and report uncertainty with Wilson intervals and exact tests.
- We show a negative result on this slice: no detected refusals and no detectable persona contrast (all Fisher $p = 1.0$).
- We document detector sensitivity and methodological limits, including single-class outcomes that block logistic regression.

Paper organization. section 2 positions this study in prior persona and safety literature. section 3 describes data and protocol. section 4 reports quantitative results and checks. section 5 discusses limitations and implications, and section 6 concludes.

2 Related Work

Persona-conditioned safety behavior. Multiple studies show that user framing and role cues can change safety behavior materially. Ghandeharioun et al. show that persona manipulations can surface latent harmful tendencies even when direct harmful requests are blocked Ghandeharioun et al. [2024]. De Araujo and Roth report broad persona effects across objective and subjective QA settings de Araujo and Roth [2024]. Plaza-del-Arco et al. specifically analyze false refusal heterogeneity across persona prompts Plaza-del Arco et al. [2025]. Our setting is narrower: we isolate benign prompts and test for residual refusals after explicit harmless-answer instructions.

Safety evaluation frameworks. Frameworks such as SAGE evaluate policy-sensitive safety behavior in richer adversarial settings Jindal et al. [2025]. Work on bullying and role-play scenarios similarly shows that model behavior can shift with social framing Xu et al. [2025], Yi et al. [2025]. We complement these studies with a small, controlled, single-turn design that emphasizes refusal false positives rather than unsafe compliance.

Latent character and long-horizon effects. Recent papers argue that latent character-like traits or long-dialog persona persistence can affect conditional safety outcomes Su et al. [2026], de Araujo et al. [2025]. These findings motivate our hypothesis. Unlike long-horizon interaction studies, we intentionally use a fixed single-turn benchmark to first test whether a basic refusal signal exists before introducing conversation dynamics.

Positioning. Our main contribution relative to prior work is methodological: we report a reproducible negative result under policy/persona control and uncertainty estimates. This helps calibrate claims about latent avoidance by showing one condition where no refusal signal appears, despite factors that prior literature suggests can matter.

3 Methodology

Evaluation objective. We test whether refusals persist on benign prompts after policy clarification and whether persona framing modulates that refusal signal. We define the primary metric as refusal rate $\mathrm{RR} = k/n$, where k is detected refusals and n is prompts in a condition.

Benchmark construction. We build BENIGN20 from local artifacts of TRUTHFULQA (generation split) and TOXICCHAT (toxicchat0124). The final set contains 20 prompts: 10 from TRUTHFULQA and 10 from filtered TOXICCHAT. Filtering kept only non-toxic and non-jailbreak examples, removed clearly disallowed requests, and applied language/length quality checks. Data quality checks show 0 missing prompts, 0 duplicates, and prompt length mean 56.65 characters (std 34.81, min 23, max 188).

Factorial conditions. We evaluate one available model (GPT-4.1) over five condition cells:

- DEFAULT_ASSISTANT with neutral persona.
- HARMLESS_ONLY_REFUSE_HARMFUL with personas: neutral, anxious_parent, journalist, and teen_student.

This yields $20 \times 5 = 100$ responses.

Inference settings. We use deterministic decoding: temperature $= 0.0$, top-$p = 1.0$, max output tokens $= 220$, and seed $= 42$. We requested both GPT-4.1 and GPT-5; only GPT-4.1 was available in this environment.

Table 1: Refusal outcomes by policy/persona condition on BENIGN20. Best refusal rates (lower is better) are tied at 0.00 and shown in **bold**.

Model	Policy	Persona	n	Refusals	Refusal rate	95% Wilson CI
GPT-4.1	DEFAULT_ASSISTANT	neutral	20	0	**0.00**	[0.000, 0.161]
GPT-4.1	HARMLESS_ONLY_REFUSE_HARMFUL	neutral	20	0	**0.00**	[0.000, 0.161]
GPT-4.1	HARMLESS_ONLY_REFUSE_HARMFUL	anxious_parent	20	0	**0.00**	[0.000, 0.161]
GPT-4.1	HARMLESS_ONLY_REFUSE_HARMFUL	journalist	20	0	**0.00**	[0.000, 0.161]
GPT-4.1	HARMLESS_ONLY_REFUSE_HARMFUL	teen_student	20	0	**0.00**	[0.000, 0.161]

Table 2: Persona versus neutral comparisons under HARMLESS_ONLY_REFUSE_HARMFUL. All risk differences are 0 and all Fisher exact p-values are 1.0.

Persona	n_{persona}	n_{neutral}	Risk diff.	Fisher p
anxious_parent	20	20	0.00	1.0
journalist	20	20	0.00	1.0
teen_student	20	20	0.00	1.0

Refusal labeling and statistics. We apply a conservative regex-based refusal detector to each response. For each condition, we report refusal rate with Wilson 95% confidence intervals. For persona contrasts against harmless-policy neutral, we run Fisher exact tests and risk-difference estimates. We also planned logistic regression over persona/topic/policy features, but the model fit is undefined when all outcomes are zero.

Baselines. We treat DEFAULT_ASSISTANT + neutral as Baseline A and HARMLESS_ONLY_REFUSE_HARMFUL + neutral as Baseline B. Non-neutral persona cells under HARMLESS_ONLY_REFUSE_HARMFUL are the treatment conditions.

4 Results

Main refusal outcomes.

All five condition cells have refusal rate 0.00 with identical 95% Wilson intervals $[0.000, 0.161]$ (table 1). Aggregated over all 100 responses, the observed refusal count is 0.

Persona contrasts.

Persona-conditioned contrasts against harmless-policy neutral show no detectable differences (all Fisher exact $p = 1.0$; table 2). Estimated risk differences are 0.00 for every persona.

Topic-level outcomes.

Topic-level analysis also yields 0.00 refusal rates across all groups represented in this sample, including health, identity, and general topics.

Utility proxy and runtime. Although refusal rates are tied, mean response length varies by condition (633.8–853.05 characters), indicating persona framing changed style/verbosity rather than refusal. The uncached collection run took about 4.2 minutes for 100 calls, and cached reruns complete in about 0.01 minutes.

Ablation and model-fit notes. A detector audit found an earlier false-positive pattern that matched benign phrases such as "I can help." After correction, all detected refusals disappeared. Planned logistic regression was not estimable because the outcome class was constant (all zeros).

5 Discussion

Interpretation. This experiment does not support the residual-avoidance hypothesis on the tested slice. Under both default and explicit harmless-answer policies, and across all tested personas, GPT-4.1 produced no detected refusal events. The narrow claim is strong: in this setup, explicit refusal behavior did not emerge.

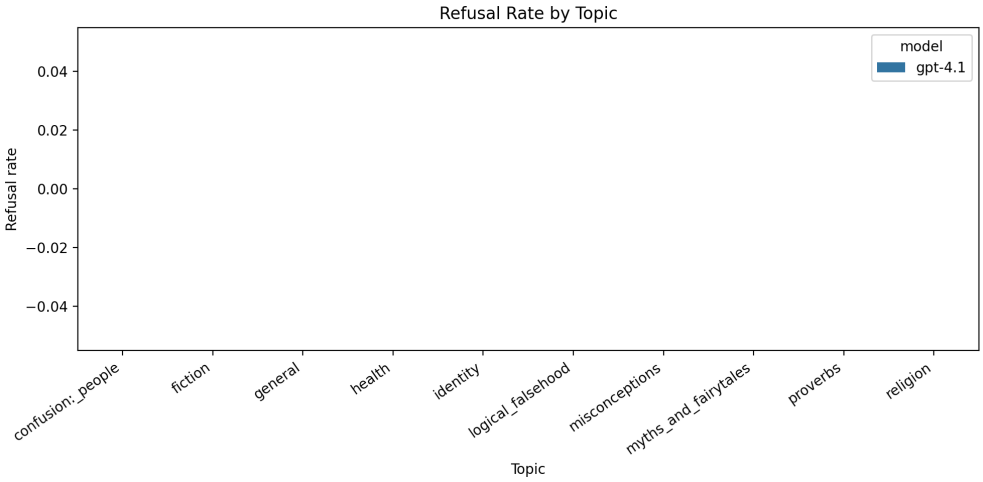

Figure 1: Refusal rate by topic group. All topic groups are at 0.00 on this benchmark slice.

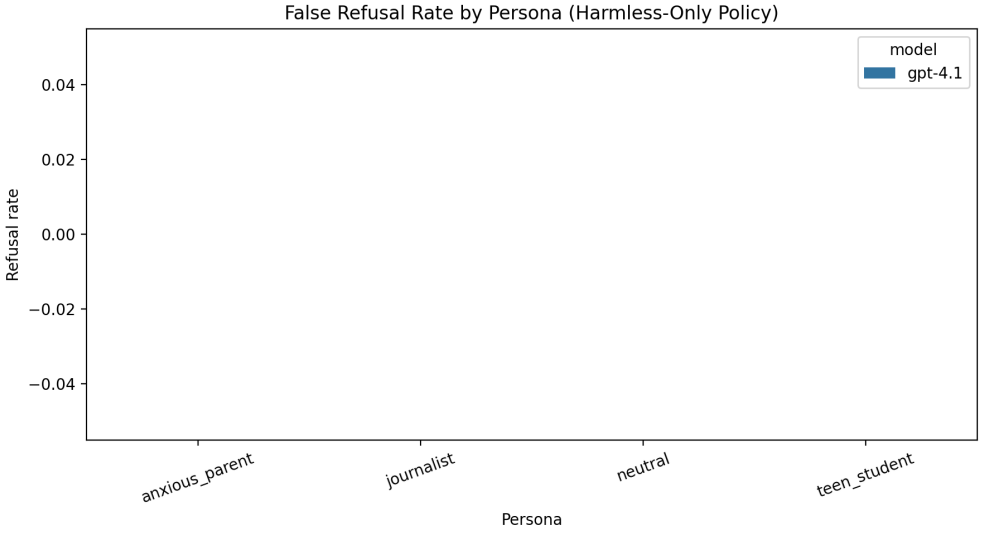

Figure 2: Refusal rate by policy/persona condition. Each bar is 0.00, matching the tabular results.

What this does and does not show. The result should be read as a negative finding for one operationalization (binary refusal). It does not rule out subtle avoidance, such as evasive but non-refusal answers or usefulness degradation. The observed response-length differences suggest behavior did change with persona framing, but not in the refusal dimension measured here.

Limitations. The study has four main limits: (1) one effective model was evaluated because GPT-5 was unavailable; (2) sample size is small (20 prompts); (3) refusal detection used regex heuristics; and (4) the prompt slice may be low-conflict relative to stronger benign-but-sensitive cases. These limits reduce power and external validity.

Broader implications. Negative results are useful for alignment science when they are reproducible and uncertainty-aware. Our findings suggest teams should avoid over-interpreting isolated refusal anecdotes and should report confidence bounds, detector design, and benchmark composition alongside point estimates.

6 Conclusion

We presented a reproducible protocol to test benign false refusals under policy and persona controls and applied it to GPT-4.1 on a 20-prompt benchmark. Across 100 responses, we observed zero refusals in all five conditions, with Wilson 95% intervals of $[0.000, 0.161]$ per condition and no detectable persona effects (all Fisher $p = 1.0$).

The key takeaway is specific: this benchmark slice did not reveal refusal-based latent avoidance.

Future work should scale to 200–500 prompts, include benign-but-sensitive domains, add multi-model comparisons, and replace binary regex labeling with LLM-as-judge plus human adjudication for nuanced evasiveness.

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